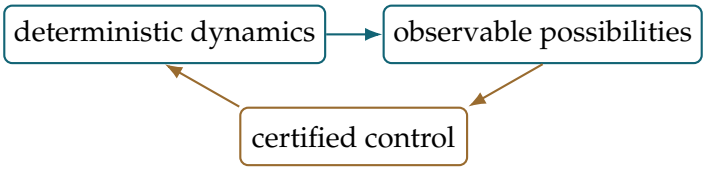
MATHEMATICAL FOUNDATIONS AND RESEARCH FRONTIERS

Determinism, Possibility, and Certified Search

A Research Monograph on Exact Interfaces, Finite Resources, and Proof-Carrying Control



DATA MIND

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Abstract

A deterministic evolution law does not, by itself, identify which observation laws it can reproduce, how much hidden memory it requires, or whether a controller can discover a desirable trajectory. This monograph develops a hierarchy of precise interfaces between deterministic maps, nondeterministic transition relations, finite-state Markov kernels, formal proofs, and generalized board games. At its core lies an exact readout/preparation intertwining and a finite-state dichotomy: the complete observable histories of a finite Markov chain admit a finite deterministic hidden-state realization in this strict sense exactly when no vertex on a directed cycle has more than one positive-probability successor. In the realizable case the minimum hidden-state cardinality is the number of admissible infinite histories. In contrast, every finite observation horizon h admits a deterministic realization; its *exact* minimum size is the number of admissible words of length $h+1$. These size results refer to freely chosen mathematical implementations, not to a predetermined physical substrate.

The second layer treats search. Nondeterministic proof-search policies generate legal alternatives; witness-bearing extensions can be encoded as legal moves of a generalized board game. An effective implementation additionally requires decidable legality and a terminating witness-discovery procedure; neither follows from abstract finite encodings alone. A given successful finite branch admits a deterministic selector, although no algorithm for finding the branch follows. A distinguished board-player can coordinate such actions only under explicitly authorized rules, and a trusted outcome requires a declared verifier. The third layer concerns theory translation: provenance-preserving records are distinct from certified proof transport, and compatible signatures do not imply either dynamic intertwining or physical equivalence. Constructions, impossibility results, simple countermodels, cost limitations, and open problems are stated separately.

The audience is a skeptical physicist with graduate mathematical maturity. No claim about fundamental indeterminism, hidden-variable ontology, universal simulation of physical theories, proof-search speedup, or experimental evidence is inferred from a formal representation theorem.

Reader's contract

The subject is deliberately narrower than the phrase *determinism versus nondeterminism* might suggest. Our claims are conditional statements about specified mathematical categories. A Markov kernel is an *observable law*, not a commitment to stochastic fundamental dynamics. A deterministic hidden map is a *representation*, not evidence that microscopic reality follows that map. A proof-producing agent is a *constrained procedure*, not a physical observer. A formal translation of derivations is not a proof that two physical theories make the same predictions.

Three distinctions govern the monograph: *possible histories versus probability-weighted histories*; *one successful history versus the family of all histories*; and *a shared notation versus a verified semantic interface*. We make the exact assumptions visible before drawing each conclusion. Speculative extensions are labeled as open problems or research proposals. The source articles' original mathematical attributions remain intact, and the collected DNI reader is treated as an editorial organization of existing material, not an extra independent theorem source. This is an integrated monograph, not a concatenation of earlier papers.

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Chapter 1

The interface problem from a physicist's viewpoint

1.1 What counts as the same dynamics?

Consider a laboratory device with a finite set Y of reportable outcomes. In discrete time one may describe its reports by a function, a set of admissible successors, or conditional transition probabilities:

$$f : Y \to Y, \qquad N : Y \to \mathcal{P}(Y) \setminus \{\varnothing\}, \qquad K : Y \to \mathcal{D}(Y). \tag{1.1}$$

These are different structures. The function assigns one next reported state to each current state. The relation assigns possibilities without weights. The kernel assigns probabilities, with $K(y,z) \geq 0$ and $\sum_z K(y,z) = 1$. In particular, the *support* of K is the possibility relation $N_K(y) = \{z : K(y,z) > 0\}$. It forgets essentially all numerical information.

If the engineer insists on deterministic hidden evolution $f : X \to X$ on a larger set X, can the observer still see the prescribed K on Y? The answer depends decisively on the meaning of *see*. Reproducing a single trajectory is weaker than reproducing every positive-probability finite history. Matching distributions at one time is weaker than matching the joint law of histories. Reproducing histories for a fixed finite observational window is weaker than reproducing them for every time. Finally, realizing the law by some freely chosen X is weaker than realizing it inside a laboratory system with a fixed phase space and allowed dynamics.

The distinction is experimentally familiar: two models can agree on a marginal distribution and disagree on correlations. Thus the observable object in this monograph is usually a *family of finite-dimensional history laws*, not one-time distributions alone.

1.2 A small obstruction with large consequences

Imagine two reportable states a and b and a Markov kernel

$$K = \begin{pmatrix} 1/2 & 1/2 \\ 0 & 1 \end{pmatrix}, \qquad Y = \{a, b\}. \tag{1.2}$$

Starting at a, the system remains at a for a geometrically distributed number of transitions and eventually reaches b almost surely. Yet there are infinitely many positive-probability observable

histories: $a, b, b, \ldots$; $a, a, b, b, \ldots$; and arbitrarily long delays before absorption. A finite hidden deterministic system, if initialized with only a finitely supported distribution over its states and allowed no subsequent randomness, produces at most finitely many different infinite reported paths. Therefore no such finite system can reproduce the entire Markov path law. This does not show that the underlying physics is nondeterministic: it shows that a particular *finite exact deterministic representation* cannot exist.

For any specified finite window, however, only finitely many observable words occur and a finite exact deterministic representation can be built. Its minimum number of hidden states will be obtained in Chapter 5. If our only goal is to reach b, a simple policy chooses $a \mapsto b$ and succeeds immediately. This third statement is about target selection, not probabilistic emulation. The tension among these three truths motivates the entire development.

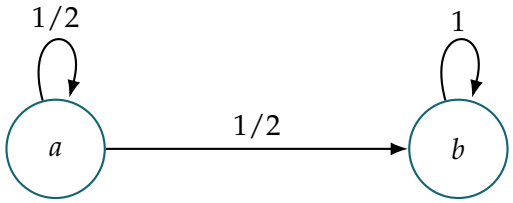


Figure 1.1: Recurrent branching at a: eventual absorption with probability one does not entail a finite exact hidden-state realization of all observable histories.

1.3 Four interface questions

The same vocabulary will be used for distinct obligations. *Dynamical realization* asks for exact equations relating readout, preparation and evolution. *Possibility preservation* asks only whether all and only admissible trajectories survive a translation. *Controlled search* asks how one permitted transition is selected toward a target. *Certified mathematical transport* asks whether a proof in one formal theory can be translated into an accepted proof in another. An answer in one category does not, without further hypotheses, answer questions in the others.

The conceptual unification is methodological, not ontological: to call two descriptions an *interface*, specify its objects, translations, diagrams, costs and acceptance obligations. Mathematical impossibility is always relative to that complete declaration.

1.4 Reading order and provenance

Chapters 2–6 develop dynamical DNI from elementary kernels through exact and finite-horizon realization, examples and costs. Chapters 7–9 introduce formal proofs, search and the board-player controller as a separate application of the interface discipline. Chapters 10–11 analyze translations between theories and the combined geometry of histories. Chapter 12 identifies the unresolved frontiers. The relationship to the four pre-existing DNI-related documents is explained precisely in Appendix B; the prior collected reader contributes organization, not an additional independent claim.

Chapter 2

Exact interfaces and observable history laws

2.1 Probability and deterministic readout

Let Y be a nonempty finite observable alphabet and X a measurable hidden-state space. For a measurable map g write $g_*\mu$ for pushforward of a probability measure. Let K be a Markov kernel on Y, $f : X \to X$ a measurable deterministic evolution, $R : X \to Y$ a readout and $E : Y \to \mathcal{D}(X)$ a preparation kernel. The composite kernel EK is

$$(EK)(y, A) = \sum_{z \in Y} K(y, z)E(z, A). \tag{2.1}$$

In physical language $E(y)$ describes how hidden initial states are prepared, conditioned on a reported initial state y. No interpretation of that preparation as ignorance or objective chance is built into its definition.

Definition 2.1 (Exact stochastic interface)**.** The quadruple (f, K, R, E) is an *exact deterministic–stochastic interface* if, for every $y \in Y$,

$$R_*E(y) = \delta_y, \qquad \text{initial readout is correct,} \tag{2.2}$$

$$f_*E(y) = \sum_{z \in Y} K(y, z)E(z), \qquad \text{preparation intertwines dynamics.} \tag{2.3}$$

These are identities of probability measures, not merely statements that the two sides have the same support.

When X is also finite, define $P(x, x') = \mathbf{1}_{\{f(x)=x'\}}$, $\Phi(x, y) = \mathbf{1}_{\{R(x)=y\}}$, and $\Lambda(y, x) = E(y, \{x\})$. Under row-vector conventions equations (2.2)–(2.3) are

$$\Lambda\Phi = I_Y, \qquad \Lambda P = K\Lambda. \tag{2.4}$$

This is a deterministic-upstairs instance of the classical intertwining framework associated with Rogers and Pitman [7]. The idea of an intertwining is not being claimed as a new construction.

Theorem 2.2 (All finite-dimensional laws)**.** *Under Definition 2.1, let* $X_0 \sim E(y)$, $X_t = f^t(X_0)$ *and* $Y_t = R(X_t)$. *Then for every* $n \geq 0$,

$$R_*f_*^nE(y) = K^n(y, \cdot), \tag{2.5}$$

and for $y_0, \dots, y_n \in Y$,

$$\Pr_y(Y_0 = y_0, \dots, Y_n = y_n) = \mathbf{1}_{\{y_0 = y\}} \prod_{t=0}^{n-1} K(y_t, y_{t+1}). \tag{2.6}$$

Thus the readout process is a Markov chain with the entire prescribed joint-history law.

Proof. Equation (2.3) gives $f_*^n E = EK^n$ by induction, after which (2.2) yields (2.5). For the stronger joint statement, the conditional hidden distribution immediately after an observed history ending at z is $E(z)$: at time zero this is preparation; if it holds at time t, equation (2.3) makes the unconditioned next hidden law $\sum_w K(z,w)E(w)$. By (2.2), the measures $E(w)$ have disjoint readout fibers $R^{-1}(w)$. Consequently the next observation equals w with probability $K(z,w)$ and the hidden conditional law, given that observation, is $E(w)$. Induct on t to obtain (2.6). □

Remark 2.3 (Why one-step marginals are insufficient). An arbitrary deterministic system might reproduce $K^n(y,\cdot)$ for each n while giving the wrong correlations among times. The intertwining identity (2.3), combined with the correctly prepared readout (2.2), supplies the stronger conditional-law structure. Later finite-horizon realizations will deliberately demand joint-history equality without claiming the stronger all-time intertwining.

2.2 Every Markov chain has an unrestricted path-space representation

Let $\mathrm{Path}_\infty(K)$ be the set of infinite sequences $(y_0, y_1, \dots)$ with $K(y_j, y_{j+1}) > 0$ for all j. This set is naturally measurable as a subset of $Y^{\mathbb{N}}$ with the product sigma-algebra. Define f as left shift, R as first-coordinate readout, and $E(y)$ as the Markov path measure conditional on first coordinate y. Then (2.2) is immediate and (2.3) is the usual disintegration of a Markov path by its next letter. This realization exists even when $\mathrm{Path}_\infty(K)$ is infinite or uncountable. The difficult issue is not whether a deterministic path shift can be written down; it is what resource constraints and independent physical requirements it can satisfy.

INTERPRETIVE BOUNDARY. The path-space representation stores an entire future history in the hidden initial condition. It is a mathematical coding, not a causal theory of where that information comes from, nor a finite-complexity simulator.

2.3 A ladder of equivalences, rather than one notion

The common phrase "the same model" hides several inequivalent predicates:

$$\begin{aligned} &\text{same entire history law} \quad \Longrightarrow \quad \text{same one-time marginals} \\ &\Longrightarrow \quad \text{same permitted observations.} \end{aligned} \tag{2.7}$$

None of the reverse implications holds without hypotheses. A selected target-reaching trajectory constitutes even less information than the complete support relation. We will treat each level separately rather than changing preservation requirements mid-proof.

Chapter 3

Possibility, support, and the price of forgetting weights

3.1 Set-valued dynamics

A possibilistic system is a nonempty-valued successor map $N : Y \to \mathcal{P}(Y) \setminus \{\varnothing\}$. Its admissible histories obey $y_{t+1} \in N(y_t)$, with no relative frequencies or probabilities specified. The exact deterministic–possibilistic analogue of Definition 2.1 has a nonempty preparation set $L(y) \subseteq X$ for each observable state. Write $g[A] = \{g(a) : a \in A\}$.

Definition 3.1 (Exact possibilistic interface)**.** The data (f, N, R, L) form an exact possibilistic interface when

$$R[L(y)] = \{y\}, \qquad f[L(y)] = \bigcup_{z \in N(y)} L(z) \quad (y \in Y). \tag{3.1}$$

The second equality includes both inclusions. There can be no hidden successors producing forbidden observable continuations, and every permitted next state must remain represented.

Proposition 3.2 (The positive-support bridge)**.** *Assume X, Y finite and (f, K, R, E) satisfies (2.2)–(2.3). Put $L_E(y) = \operatorname{supp} E(y)$ and $N_K(y) = \{z : K(y, z) > 0\}$. Then (f, N_K, R, L_E) satisfies (3.1).*

Proof. The first equation follows from $R_* E(y) = \delta_y$. On finite spaces, $\operatorname{supp}(f_* \mu) = f[\operatorname{supp} \mu]$ and the support of a nonnegative weighted sum of finitely many measures is the union of their supports over strictly positive weights. Taking supports of (2.3) therefore yields $f[L_E(y)] = \bigcup_{z \in N_K(y)} L_E(z)$. □

Proposition 3.3 (Support respects finite composition)**.** *For finite stochastic kernels $K_1 : Y \to \mathcal{D}(Z)$ and $K_2 : Z \to \mathcal{D}(W)$, the support relation of their composite is the relational composite of their supports:*

$$N_{K_1 K_2}(y) = \bigcup_{z \in N_{K_1}(y)} N_{K_2}(z). \tag{3.2}$$

Proof. The (y, w) entry of $K_1 K_2$ is a finite sum $\sum_z K_1(y, z) K_2(z, w)$ of nonnegative numbers. It is positive exactly when one term is positive, which is equivalent to the existence of z with both allowed support edges. □

3.2 A direct path-space representation for possibility

Take $X = \mathrm{Path}_\infty(N)$, f the shift and $L(y)$ the set of legal paths beginning at y. The left shift of $L(y)$ is exactly $\bigcup_{z \in N(y)} L(z)$; thus every finite total successor relation has an exact possibilistic interface, possibly with an infinite hidden space. As before, the question of *finite* realization is a separate question.

3.3 Forgetting is not reconstruction

The map $K \mapsto N_K$ is many-to-one. On $Y = \{0, 1\}$, the stochastic matrices

$$K_1 = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}, \qquad K_2 = \begin{pmatrix} 9/10 & 1/10 \\ 1/3 & 2/3 \end{pmatrix}$$

have the same complete positive-support graph but different path probabilities. Any theorem stated only in terms of positive successors can apply to both; a theorem about exact stochastic laws cannot be recovered from their common support without additional data.

Remark 3.4 (Multiple move labels). A transition graph records *successor states*. A proof search or board game may permit two distinct move labels producing the same successor. Projecting away the labels erases the distinction between those moves. When a representation theorem is about state words, the word count must not silently be replaced by the number of move-labeled histories. To preserve labels, enlarge the observable alphabet to encode them and re-establish the hypotheses for that enlarged process.

3.4 A categorical viewpoint, with limits

There is a useful discipline here: deterministic functions embed into Markov kernels by Dirac distributions, and positive support maps finite kernels to total relations in a composition-respecting manner. These constructions motivate the language of functorial interfaces. More general theories of stochastic morphisms, including Markov categories [9], make such relationships precise. But neither a diagram of categories nor a support map automatically supplies a resource-optimal hidden dynamics, a proof translator, or an equivalence of physical theories. The existence of a structure-preserving mapping must always be separated from the properties one hopes that mapping preserves.

Chapter 4

The finite exact-realization obstruction

4.1 Positive-transition graph and history counts

Assume henceforth in this chapter that Y is finite, nonempty, and every vertex has at least one outgoing legal successor. Let $G = (Y, A)$ be the digraph with $(y, z) \in A$ exactly when $z \in N(y)$. The stochastic case is recovered by taking $N = N_K$. Denote by $\mathrm{Word}_n(N)$ the set of admissible *state* words $(y_0, \dots, y_{n-1})$ of length n, counting different starting states separately, and write

$$B_n(N) = |\mathrm{Word}_n(N)|, \qquad \mathrm{Path}_\infty(N) = \{\text{all legal infinite state paths}\}. \tag{4.1}$$

A legal infinite path is one whose *every finite transition* has positive conditional probability in the Markov case. This does not imply the singleton infinite path has positive probability: for example, the infinite all-zero history in an i.i.d. fair binary process is legal, but has probability $\lim_{n\to\infty} 2^{-n} = 0$. All lower bounds below use positive probability of *finite* words, not of individual infinite paths.

A vertex is *cyclic* if it lies on a directed cycle, including a self-loop. A cycle vertex with at least two distinct successors is a *recurrent branch*. This term is graph-theoretic; it does not assert recurrence of a stochastic sample path with probability one.

Lemma 4.1 (Word-count lower bound)**.** *If a finite deterministic hidden state space X realizes all admissible N-histories by the exact equations (3.1), then $|X| \geq B_n(N)$ for every $n \geq 1$. The same lower bound holds for finite exact stochastic interfaces with support N_K.*

Proof. An initial hidden state x determines exactly one observable word $R(x), R(fx), \dots, R(f^{n-1}x)$. The set identities (3.1) imply by induction that every admissible length-n word starting at y is realized by some $x \in L(y)$. Distinct words require distinct hidden initial states. In the stochastic case every legal word has positive probability under the Markov law, by Theorem 2.2, so some positively weighted hidden state must realize it. In either case at least B_n hidden states are necessary. □

Lemma 4.2 (Graph-path finiteness criterion)**.** *For a finite total digraph, the following are equivalent:*

(i) the set $\mathrm{Path}_\infty(N)$ is finite;

(ii) the sequence $B_n(N)$ is bounded uniformly in n;

(iii) every cyclic vertex has exactly one successor.

Proof. Every finite word extends to an infinite path because the relation is total. If the infinite-path set is finite, every level of finite prefixes has at most its cardinality, proving (i)⇒(ii).

Suppose a cycle passes through v and v has another successor w distinct from the successor used by that cycle. One can traverse the cycle j times and then take the edge $v \to w$, for each $j \geq 0$. These paths can be extended indefinitely. Their prefixes are pairwise distinguishable by sufficiently long lengths, so B_n cannot remain bounded. Thus (ii)⇒(iii). Here it does not matter whether the alternative successor eventually returns to the cycle: the first deviation time distinguishes the paths.

Conversely, suppose (iii). Any infinite path in a finite graph must repeat a vertex. Once it first enters a directed cycle, every vertex on that cycle has exactly one successor; hence all subsequent choices are forced. The entire infinite path is determined by its initial segment before the first repeated vertex, whose length is bounded in terms of $|Y|$. There are only finitely many such initial segments, proving (iii)⇒(i). □

Theorem 4.3 (Exact finite deterministic realization dichotomy)**.** *Let K be a finite-state Markov kernel and $N = N_K$. The following conditions are equivalent:*

(a) there exist finite X, deterministic $f : X \to X$, readout R, and preparation kernel E satisfying (2.2)–(2.3);

(b) every vertex lying on a directed cycle in G_N has exactly one positive-probability successor;

(c) $\sup_{n\geq 1} B_n(N) < \infty$.

When these hold, the minimum hidden-state cardinality is

$$m_\infty(K) := \min |X| = |\operatorname{Path}_\infty(N)| = \sup_{n\geq 1} B_n(N). \tag{4.2}$$

If they fail there is no finite exact stochastic interface of this kind. The same existence criterion and cardinality formula hold for finite total possibilistic systems under (3.1).

Proof. If a finite exact interface exists, Lemma 4.1 bounds B_n by $|X|$ for every n; Lemma 4.2 yields (b). Conversely assume (b). Then $X = \operatorname{Path}_\infty(N)$ is finite. Use the left shift f, first-coordinate readout R, and for $E(y)$ the probability of each admissible infinite path under the Markov law started at y. This is a probability distribution on a finite set: every legal infinite path has an eventually forced cycle, and the Markov law is concentrated on precisely these paths. Disintegrating by the first successor gives $f_*E(y) = \sum_z K(y,z)E(z)$, and initial readout gives $R_*E(y) = \delta_y$. The possibilistic case uses $L(y)$ equal to all paths starting at y.

The construction uses $|\operatorname{Path}_\infty(N)|$ hidden states. Conversely, because that set is finite, there is one common n_0 for which the length-n_0 prefixes of its distinct paths are all distinct. Therefore $B_{n_0} = |\operatorname{Path}_\infty(N)|$ and Lemma 4.1 proves that no smaller finite X is possible. Prefix counts are nondecreasing, since all prefixes extend, so the supremum equals this value. □

4.2 An effective count without enumerating hidden systems

When the criterion holds, let $c(y)$ denote the number of legal infinite paths starting at y. If y lies on a cycle, then $c(y) = 1$. If y is not cyclic, set

$$c(y) = \sum_{z\in N(y)} c(z), \qquad m_\infty = \sum_{y\in Y} c(y). \tag{4.3}$$

The subgraph induced by noncyclic vertices is acyclic, so this recursion is well-founded and can be evaluated in reverse topological order after computing strongly connected components. The count can be exponentially large in the bit length of the graph description, but its defining recurrence is finite and exact. Detecting the obstructing recurrent branch requires no enumeration of hidden maps.

4.3 What the obstruction does not say

Theorem 4.3 permits free choice of f, R and E, but it forbids an infinite number of discrete hidden states. It neither constrains preparation precision nor tests a prescribed physical dynamics. Its conclusion also does not rule out infinite-dimensional or continuum-valued deterministic path representations. Nor does it speak directly to an infinite, stage-dependent search state (p, m): the variable m makes a nominally finite graph infinite unless one proves a finite time-homogeneous quotient with the correct interface properties.

INTERPRETIVE BOUNDARY. The words "no finite hidden representation" are not equivalent to "irreducible physical chance" or to "computational nondeterminism is necessary." The theorem is a structural cardinality obstruction in a fixed mathematical interface class.

Chapter 5

The sharp cost of a finite observation horizon

5.1 Finite observational equivalence

For a fixed integer $h \geq 0$, experimental observation can access only $h+1$ recorded values $Y_0, \ldots, Y_h$. It is then mathematically natural to require exact reproduction of *their joint distribution*, without demanding that the same hidden state law intertwine the process for all later times. This weaker problem is not merely a cutoff approximation: it has an exact minimum-size theorem.

Definition 5.1 (Horizon-h deterministic realization)**.** An h-history realization of a finite Markov kernel K comprises finite X, one time-homogeneous deterministic self-map $f : X \to X$, readout $R : X \to Y$, and initial preparations $E(y)$, such that for every initial observable state y the law of

$$\big(R(X_0), R(fX_0), \ldots, R(f^h X_0)\big), \quad X_0 \sim E(y) \tag{5.1}$$

coincides *jointly* with the first $h+1$ observations of the K-chain from y. Denote by $m_h(K)$ the minimum possible $|X|$. No all-time identity (2.3) is required.

Theorem 5.2 (Sharp finite-horizon realization)**.** *For every stochastic kernel K on nonempty finite Y and $h \geq 0$,*

$$\boxed{m_h(K) = B_{h+1}(N_K) = |\operatorname{Word}_{h+1}(N_K)|.} \tag{5.2}$$

Thus every finite window admits an exact time-homogeneous *deterministic hidden-state realization, even when Theorem 4.3 rules out a finite realization for all times.*

Proof. For the lower bound, every initial hidden x determines one length-$(h+1)$ observation word. Every admissible word $(y_0, \ldots, y_h)$ has positive probability under the Markov law from y_0 and therefore must be generated by at least one hidden state with positive preparatory probability. Distinct words need distinct initial hidden states, hence $|X| \geq B_{h+1}$.

For the upper bound define $X = \operatorname{Word}_{h+1}(N_K)$. Since each row of K is stochastic, choose once and for all an allowed successor $a(z) \in N_K(z)$ for every $z \in Y$. For each $w = (y_0, \ldots, y_h) \in X$, set

$$R(w) = y_0, \qquad f(w) = (y_1, \ldots, y_h, a(y_h)). \tag{5.3}$$

For $h = 0$ the last expression is interpreted as the singleton $(a(y_0))$. The word $f(w)$ remains admissible, so $f : X \to X$ is total and independent of time. Define

$$E(y)(w) = \mathbf{1}_{\{y_0=y\}} \prod_{j=0}^{h-1} K(y_j, y_{j+1}), \tag{5.4}$$

with the empty product equal to one. By summing over all legal continuations from y, these weights total one. For $0 \le j \le h$, left shifts reveal $R(f^j w) = y_j$. Therefore the joint observation law generated by $E(y)$ agrees exactly with the Markov law for all required steps. The construction has precisely B_{h+1} hidden states and attains the lower bound. □

Corollary 5.3 (Possibilistic finite-horizon minimum)**.** *If $N : Y \to \mathcal{P}(Y) \setminus \{\varnothing\}$ is finite and $m_h(N)$ counts the least hidden states for which deterministic dynamics reproduce* exactly the set *of legal length-$(h+1)$ state words from every initial state, then*

$$m_h(N) = |\,\mathrm{Word}_{h+1}(N)|.$$

Proof. Each hidden initial state produces only one observed word, so every admissible word requires a distinct hidden initial state. Conversely use the shift construction (5.3) on legal words and take $L(y)$ to be all words beginning at y. □

5.2 Information content and the distinction from runtime

Identifying one of $m_h(K)$ hidden states requires at least $\lceil \log_2 m_h(K) \rceil$ binary bits in a lossless fixed-length code. Thus $\log_2 B_{h+1}$ is an elementary *discrete-state capacity* lower bound. It is not a lower bound on machine instruction count, memory bandwidth, inference calls, or randomness generation. Preparation probabilities may require arbitrary numerical precision. The finite-horizon theorem optimizes only the number of abstract hidden states over all constructions.

5.3 The combinatorics of growth

Let A be the 0–1 support adjacency matrix, with $A_{yz} = 1$ exactly when $z \in N(y)$. Writing $\mathbf{1}$ for the all-ones column, count admissible words by matrix multiplication:

$$B_{h+1}(N) = \mathbf{1}^{\mathsf{T}} A^h \mathbf{1}. \tag{5.5}$$

Indeed, each term in A^h counts directed paths of h transitions, and summation over endpoints and initial states counts every admissible word once. Combined with Theorem 5.2, this supplies an exact linear-algebraic expression for the finite-horizon hidden-state minimum.

Corollary 5.4 (No universal exponential lower rate)**.** *If the graph contains recurrent branching, then $m_h \to \infty$, but its divergence need not be exponential. Both linear and exponential rates arise from two-state total graphs.*

Proof. Theorem 4.3 and monotonicity of B_n establish divergence. For the graph $a \to \{a, b\}$, $b \to b$, there are $h + 1$ words beginning at a and one beginning at b, so $m_h = h + 2$. For the complete two-state graph every binary word is permitted, so $m_h = 2^{h+1}$. □

Remark 5.5 (Finite windows do not form one uniform finite simulator). The optimal hidden space X_h changes with h. It is invalid to fix some finite h, exhibit a finite exact realization on that window and infer a single finite exact realization valid for all h. Recurrent branching forces $|X_h| \to \infty$ even though each individual X_h is finite.

5.4 An experimentally testable restriction—and its boundary

If a physical modeling proposal specifies an upper budget of M discrete hidden states, exact reproduction of all h-step histories in this interface class requires $B_{h+1} \leq M$. Violating the inequality excludes that *model class*. It does not exclude systems with continuous hidden variables, measurements that disturb evolution, time-dependent control, contextual response functions, or different admissible preparation rules. Scientific use of the theorem demands that those restrictions be independently justified rather than smuggled in as "determinism."

Chapter 6

Fixed implementations, approximation, and worked models

6.1 An optimal four-state hidden model

Let $Y = \{a, b, c\}$ and define $K(a, b) = 1/3$, $K(a, c) = 2/3$, $K(b, b) = K(c, c) = 1$, with all other entries zero. The legal infinite paths are

$$w_1 = ab^\infty, \quad w_2 = ac^\infty, \quad w_3 = b^\infty, \quad w_4 = c^\infty.$$

Take $X = \{w_1, w_2, w_3, w_4\}$; under left shift $f(w_1) = w_3$, $f(w_2) = w_4$, and $f(w_3) = w_3$, $f(w_4) = w_4$. Read the initial symbol, and put $E(a) = \frac{1}{3}\delta_{w_1} + \frac{2}{3}\delta_{w_2}$, $E(b) = \delta_{w_3}$ and $E(c) = \delta_{w_4}$. Direct calculation gives

$$f_*E(a) = \tfrac{1}{3}E(b) + \tfrac{2}{3}E(c), \quad f_*E(b) = E(b), \quad f_*E(c) = E(c),$$

so the intertwining holds. All four infinite paths have distinct initial words of some finite length; therefore four hidden states are *necessary*, not just sufficient. The number four is a cardinality optimum, not an implementation cost in physical units.

6.2 Almost-sure absorption is not all-history finite realizability

For the two-state kernel (1.2), starting from a gives $\Pr(Y_t = a) = 2^{-t}$ and eventual absorption at b almost surely. Nonetheless every delay length $j \geq 0$ has positive probability $2^{-(j+1)}$. No finite hidden model of the exact all-history form exists. On the horizon h, its exact minimum is $h + 2$. For the complete two-state kernel with every transition probability $1/2$, the horizon minimum is 2^{h+1} and the all-history finite obstruction likewise applies. "Almost surely" and "all admissible positive-probability histories" are not interchangeable quantifiers.

6.3 What happens when the hidden machine is prescribed?

The previous chapters optimized over freely chosen f and R. Suppose instead an apparatus with a specified finite hidden set X, prescribed deterministic transition f and fixed readout R is handed to us. Only the preparation kernel E may be designed. This is a more physically meaningful, and distinct, feasibility question.

Proposition 6.1 (Fixed-model exact compatibility is linear feasibility)**.** *Given finite X, Y, prescribed $f : X \to X$, $R : X \to Y$, and stochastic K, let e_{yx} be unknown reals. An exact interface exists if and only if the following system is feasible:*

$$e_{yx} \geq 0, \qquad \sum_{x \in X} e_{yx} = 1 \qquad (y \in Y), \tag{6.1}$$

$$e_{yx} = 0 \qquad \text{when } R(x) \neq y, \tag{6.2}$$

$$\sum_{u : f(u) = x} e_{yu} = \sum_{z \in Y} K(y, z) e_{zx} \qquad (y \in Y,\ x \in X). \tag{6.3}$$

If K has rational entries, every feasible system of this form has a rational solution.

Proof. The variables e_{yx} are exactly $E(y, \{x\})$. Nonnegativity and normalization define probabilities, (6.2) is equivalent to (2.2), and the coordinates of (6.3) express (2.3) at each hidden point. If K is rational, the feasible set is a nonempty bounded polyhedron cut out by rational inequalities and equations, hence contains a rational point. □

This converts a *fixed prescribed source* problem into a linear program; it does not determine the least cardinality over additional design constraints or imply that a particular Hamiltonian flow and measurement protocol can be represented by a finite state automaton.

6.4 Approximate interfaces

Exactness is a strong idealization, especially when physical parameters are only measured approximately. Suppose (2.2) is exact but (2.3) has error ε in total variation:

$$\sup_{y \in Y} \left\| f_* E(y) - EK(y) \right\|_{\mathrm{TV}} \leq \varepsilon, \quad \|\mu - \nu\|_{\mathrm{TV}} := \sup_A |\mu(A) - \nu(A)|.$$

Theorem 6.2 (Linear marginal-error accumulation)**.** *For every $n \geq 0$,*

$$\sup_{y \in Y} \left\| R_* f_*^n E(y) - K^n(y, \cdot) \right\|_{\mathrm{TV}} \leq \min\{1, n\varepsilon\}. \tag{6.4}$$

Proof. Define $d_n = \sup_y \|f_*^n E(y) - EK^n(y)\|_{\mathrm{TV}}$. We have $d_0 = 0$. Insert and subtract $f_*^n EK(y)$ into the difference defining d_{n+1}. Deterministic pushforwards contract total variation, and mixing by the Markov kernel K does not amplify a uniform total-variation error. Therefore $d_{n+1} \leq d_n + \varepsilon$, so $d_n \leq n\varepsilon$. Readout also contracts total variation; since $R_* EK^n = K^n$ by (2.2), the claim follows. Total variation is at most one. □

Remark 6.3 (The error bound is marginal, not a path-law certificate). Equation (6.4) bounds the distribution of the *single observation at time n*. It does not, without a separate argument, bound the total variation of the joint history $(Y_0, \ldots, Y_n)$ by $n\varepsilon$. Claiming such a stronger result solely from the displayed estimate would be a mathematical error. Approximate full-history interfaces deserve their own definition and analysis.

6.5 A hierarchy of resource questions

At least six different budgets are possible: hidden-state cardinality; number of bits needed to encode a hidden state; numerical precision and entropy of preparation; time to compute one f-step; cost of synthesizing f, R, E; and observational or verification overhead. A theorem optimizing the first quantity is not an asymptotic speedup theorem for the others. The same warning will reappear when abstract proof steps are compiled into GBG moves and charged as computation.

Chapter 7

Nondeterministic proof search as a distinct dynamical system

7.1 Proof states and legal extensions

Fix a finitary formal inference system F over a set of formulas Y. Let $\Gamma \subseteq Y$ be a *finite* hypothesis set and $\mathrm{Con}_F(\Gamma)$ the set of derivable formulas. A finite proof state p stores formulas already available in a search. Infinite premise sets require a separate finite-state encoding discipline and are not assumed in the compilation arguments here. Inference rules supply *direct consequences* of p; every proof step is required to preserve sound derivability. This is not a probabilistic Markov process unless a probability model is additionally specified.

Following *Depths of Induction* v49, Section 5 [5], a nondeterministic proof-search policy is a nonempty-valued mapping

$$\Sigma(p, m) \subseteq \{\Delta \subseteq_{\mathrm{fin}} Y : \text{each formula in } \Delta \text{ is a direct consequence of } p\}, \quad \Sigma(p, m) \neq \varnothing, \tag{7.1}$$

where $m \geq 1$ is the stage. A branch is a sequence

$$p_1 = \Gamma, \qquad p_{m+1} = p_m \cup \Delta_m, \qquad \Delta_m \in \Sigma(p_m, m). \tag{7.2}$$

The word *nondeterministic* refers to the set of permitted choices, not to quantum indeterminacy, stochastic probabilities, or an ability to know in advance which choice is productive. Singleton-valued policies are valid special cases.

Proposition 7.1 (Soundness of a permitted branch)**.** *If $\Gamma \subseteq \mathrm{Con}_F(\Gamma)$, and every admitted extension contains only genuine direct consequences of the current proof state, then $p_m \subseteq \mathrm{Con}_F(\Gamma)$ for every $m \geq 1$.*

Proof. Induct on m. The initial state consists of hypotheses, and $\mathrm{Con}_F(\Gamma)$ is closed under valid inference. Consequently adjoining genuine direct consequences of p_m preserves membership in $\mathrm{Con}_F(\Gamma)$. □

This is the essential soundness component of the branch-SIC treatment in the source paper. It explains why following a legal branch never creates an underivable proposition. It does not say a given branch reaches every derivable goal.

7.2 Covering proofs versus discovering proofs

One useful strength condition is *proof covering*: for every finite proof from Γ there is a branch whose accumulated formulas contain the successive proof prefixes. If a target T has a finite proof and such a policy is proof-covering, at least one branch reaches T. Indexing requires care: with $p_1 = \Gamma$, the first extension produces p_2. Under the shifted-prefix convention $\{\pi(1), \ldots, \pi(j)\} \subseteq p_{j+1}$, tracking a proof of length n reaches its last formula by stage $n + 1$; a stage-n conclusion needs the stronger convention that the first proof formula is already present in p_1. We use only the qualitative finite-reachability consequence, not an unqualified numerical horizon. The statement is existential in the branch, and proof length is not identical to runtime or aggregate search cost.

Theorem 7.2 (Deterministic witness selection from a given trace)**.** *Suppose a stage-complete search state is (p, m) and every legal extension at every intended nonterminal state has an available finite accepted witness. If a finite legal branch reaches T, there exists a deterministic (possibly nonuniform) selector whose run from the same initial configuration reaches T along that branch. If enabled witnesses are effectively selectable off the trace, a supplied finite successful trace yields a computable selector. Neither conclusion gives a method to* find *the successful trace.*

Proof. On each configuration (p_j, j) of a given successful trace, choose the already recorded extension and an accepted witness. The stage coordinates make the listed configurations distinct even if $p_j = p_{j+1}$, so the prescribed choices are consistent. Choose any available enabled witness at all other nonterminal states. Witness coverage ensures such choices exist. The resulting deterministic selector reproduces the trace from the initial configuration by induction. If the off-trace witness choice is effective, the finite table of trace choices can be hard-coded, giving a computable selector conditional on the supplied table. Constructing the table from the original proof-search instance, however, may require solving the original search problem. □

Remark 7.3 (Do not erase the clock). The mapping $(p, m) \mapsto p$ is not necessarily a valid transition-system quotient when Σ depends on m. Two identical proof sets at different stages may have different enabled extensions. In particular, a stage-complete space may be infinite even with finitely many underlying formulas. The finite-state realization theorem of Chapter 4 therefore cannot be applied merely by ignoring stage indices.

7.3 Search effectiveness is a stronger quantifier statement

Let I be an input instance and $\mathsf{Sol}(I)$ its successful traces. Theorem 7.2 has the form

$$\forall I \, \big(\mathsf{Sol}(I) \neq \varnothing \implies \exists \delta_I \; \delta_I \text{ reaches the target}\big).$$

A uniform efficient search theorem would require an effectively computable map $I \mapsto \delta_I$, with a proved resource bound including the work of computing that map. Moving an existential witness inside an algorithmic construction is a substantive additional obligation. It is not delivered by the mere fact that one successful branch exists.

7.4 The target-search separation

Take successor states a, b with $N(a) = \{a, b\}$, $N(b) = \{b\}$ and target b. The deterministic choice $a \mapsto b$ reaches the target in one move. Nevertheless the full infinite history relation has recurrent

branching and no finite exact all-history realization. If instead an adversary selects the edge from a, the adversary may keep choosing $a \to a$ and prevent a forced win. Thus a single graph separates three propositions: successful deterministic selection exists; full finite exact realization is impossible; adversarially guaranteed reachability fails. These are not contradictory because their quantifiers and preservation requirements differ.

Chapter 8

Generalized board games as proof-carrying transition systems

8.1 Turning permissible inference into a legal move

In the generalized-board-game (GBG) framework [6], a complete game state is a formal object whose encoding contains everything relevant to future legal transitions: visible location reports, current player, stage counters, terminal status, and any other rule-dependent metadata. For a canonical move label μ let r_μ be a partial unary map on the set S of rule-complete states. The move μ is legal at s precisely when $r_\mu(s)$ is defined. The dictionary

$$\text{legal canonical move label } \mu \quad \longleftrightarrow \quad \text{partial unary inference rule } r_\mu \tag{8.1}$$

is part of the formal construction. Reachability from s_0 by finitely many move applications corresponds to derivability from $\{s_0\}$ in that unary-rule system. A move-labeled path is simultaneously a legal match and a match-proof. These statements are formal encodings; they are not claims that every proof system is literally a competitive board game.

8.2 Accepted witnesses are an extra datum

The central subtlety is that admitting an extension $\Delta \in \Sigma(p, m)$ mathematically is not the same as possessing an executable move label that a checker accepts. For each admissible extension specify a set $\mathcal{W}(p, m, \Delta)$ of *accepted, finitely encoded witnesses*. The intended compilation must require

$$\forall (p, m) \text{ nonterminal}, \quad \forall \Delta \in \Sigma(p, m), \quad \mathcal{W}(p, m, \Delta) \neq \varnothing. \tag{8.2}$$

Without this hypothesis a mathematically permitted proof step may be unavailable to the implemented game, breaking the claimed correspondence of search branches and matches.

Theorem 8.1 (Witness-complete search compilation)**.** *Suppose all intended reachable configurations (p, m) and labels (Δ, w) admit finite-string encodings, every accepted witness asserts a sound direct extension, and (8.2) holds. Associate to the label $\mu = (\Delta, w)$ the partial rule*

$$r_\mu(p, m) = (p \cup \Delta, m + 1) \tag{8.3}$$

when $\Delta \in \Sigma(p, m)$ and $w \in \mathcal{W}(p, m, \Delta)$. Then the resulting rule-complete transition system is a GBG in the unary-rule sense. Its finite move-labeled match-proofs are in bijection with finite witness-labeled search branches; forgetting witnesses is a surjection onto the admitted finite Σ-branches. Every accumulated proof state remains within $\mathrm{Con}_F(\Gamma)$.

Proof. The complete state specifies which labels are enabled and their unique successors. Give every canonical label its own partial unary rule; labels remain distinct even if they produce equal successors. Applying a rule is exactly taking the corresponding witness-labeled extension, so the label-sequence-to-branch correspondence and its inverse follow inductively. By (8.2), every permitted unlabeled extension has at least one labeled representative; forgetting witnesses is therefore onto, but may not be injective. Soundness follows from Proposition 7.1. □

Remark 8.2 (Computability is not automatic). The theorem defines an *abstract* rule system over finite encodings. Encodability and nonempty witness sets do not imply that membership in $\Sigma(p, m)$ is decidable, that a purported certificate can be checked effectively, or that an enabled witness can be found in finite time by a given procedure. Distinct accepted witnesses for one extension create distinct labels but not necessarily distinct observable successor states.

Corollary 8.3 (Effective witness-complete compilation)**.** *In addition to the hypotheses of Theorem 8.1, assume that reachable configurations and candidate labels have computable finite encodings and decoding procedures, that a* frozen total computable *predicate*

$$\mathrm{Legal}(p, m, \Delta, w) = \big[\Delta \in \Sigma(p, m) \ \wedge \ V(p, m, \Delta, w) = \mathrm{accept}\big]$$

decides legal labeled extensions, and that acceptance by V certifies direct consequence from the current proof state under the declared formal system. Assume that each reachable nonterminal configuration has an accepted labeled extension. Then the compiled GBG has an effective successor test, and a computable (not necessarily efficient) procedure can select one enabled move at any given nonterminal configuration. If a finite successful trace is additionally supplied, a computable selector can follow that trace.

Proof. The total predicate decides rule applicability and the successor is computed by $(p, m) \mapsto (p \cup \Delta, m + 1)$. Enumerate candidate finite-string labels in a fixed effective order and test them using Legal. Since at least one legal label exists, the enumeration halts and returns an enabled move. For a supplied successful trace, replace this choice by the recorded accepted labels at its finitely many stage-indexed configurations, using the effective default elsewhere. Acceptance soundness and Theorem 8.1 ensure that all resulting accumulated states remain derivable. No upper bound on the search through labels follows, and no successful trace is constructed from the instance alone. □

Remark 8.4 (Scope of effectivity). The predicate V is a verifier for the *specified syntactic rule and witness calculus*, not an oracle for semantic truth. If the condition $\Delta \in \Sigma(p, m)$ is undecidable, or the verifier is only partially defined, the corollary does not apply even if every legal step has some mathematical witness. A compiled, executable legal-move mechanism is still not a general-purpose theorem-discovery algorithm.

8.3 A refinement map and its limitations

The map $q(p, m) = p$ forgets the stage, and may be useful as a *report*. It need not preserve all enabled transitions. Similarly, a map that forgets witness labels may preserve state reachability

but not proof identity. A certificate-producing simulation must specify not only a state projection but a transformation from accepted upstairs derivations to accepted downstairs derivations. The map of states alone proves no such fact.

8.4 History counts and inference costs

If $\mathsf{M}_n(s)$ counts the move-labeled matches of length n from state s and every state's enabled move set is finite, then

$$\mathsf{M}_0(s) = 1, \qquad \mathsf{M}_{n+1}(s) = \sum_{\mu \text{ enabled at } s} \mathsf{M}_n(r_\mu(s)). \tag{8.4}$$

This follows by partitioning histories according to their first move. But M_n counts move *histories*; B_{n+1} from the finite realization theorem counts observable *state words*. Two distinct labels can lead to the same state, and an accepted macro-extension can contain several original inferences. Thus match depth, source proof length, total inference count, and charged verifier time must not be substituted for each other in quantitative statements.

Chapter 9

Controller authority, adversarial forcing, and certification

9.1 The board-player is not an oracle

The GBG+ framework augments a GBG with a distinguished board-player $P_0 = B^\star$. Its *base* form does not supply all conceivable scheduling, reflection, mutation, or certificate powers. Those are optional, explicitly declared packages. An instruction that the board may select an enabled participant is not, by itself, permission to select that participant's particular mathematical inference. Similarly, scheduling a prover is not verifying its conclusion.

Definition 9.1 (Authorized search-selection package)**.** An active GBG+ control package specifies: (i) what the board may read about the ordinary search state; (ii) which witness-labeled extension choices it may submit or dispatch; (iii) a legal commitment rule that performs the ordinary transition; (iv) a projection erasing purely administrative actions; and (v) any verifier and cost-accounting rules. No authority beyond these declarations is assumed.

Proposition 9.2 (Stage-faithful conditional board-controller simulation)**.** *Let S^+ be the configurations of an active GBG+ extension, and let $q : S^+ \to S$ project onto a space S of* complete *ordinary configurations (p, m), including the current stage m. Fix a deterministic ordinary selector $\delta(p, m) = (\Delta, w)$ choosing a legal witness-labeled extension at each nonterminal configuration of interest. Assume:*

(i) the initial augmented configuration s_1 satisfies $q(s_1) = (\Gamma, 1)$;

(ii) the board has explicit legal authority *to select and commit $\delta(p, m)$ when the projected state is (p, m);*

(iii) every administrative action is q-stuttering: $q(s') = q(s)$;

(iv) each commitment of $\delta(p, m)$ has exactly the projected effect

$$q(s') = (p \cup \Delta, m + 1) \qquad \text{when } q(s) = (p, m),$$

with no other non-stuttering projected actions; and

(v) for each ordinary step being simulated, the controller execution completes a finite *legal block of zero or more administrative moves followed by the stipulated commitment.*

Then the q-projection of every such compliant execution, after deletion of administrative stutters, is exactly the ordinary stage-indexed run of δ from $(\Gamma, 1)$. If a fixed sound verifier for the frozen formal system accepts every trusted deposit as a proof judgment under the unchanged hypotheses, all such deposited statements are derivable in that system.

Proof. By (i) both traces have the same complete initial configuration. Assume their configurations agree after j commitments and equal (p, m). Conditions (iii) and (v) imply that the intervening finite administrative prefix leaves (p, m) unchanged. Conditions (ii) and (iv) ensure the next non-stuttering action is the legal commitment of $\delta(p, m)$ and takes the projection to $(p \cup \Delta, m + 1)$, which is precisely the ordinary successor. By induction, the projected states and their stages agree at every commitment. Condition (iv) excludes extraneous projected transitions. This is *trace preservation conditional on completing the prescribed blocks*; arbitrary unfair runs with infinite administrative stuttering need not make progress. For deposited judgments, acceptance by the assumed sound frozen verifier implies derivability under its declared hypotheses and inference rules. No physical realization, uniform method of finding δ, or independent checker implementation is inferred. □

Remark 9.3 (Why the stage cannot be erased). Preserving only the formula-set update is insufficient: $\Sigma(p, m)$ can allow different actions at stages m and $m + 1$ even when p is unchanged. In particular, treating an administrative scheduling move as if it advanced the ordinary proof-search clock would invalidate the claimed simulation. The full-state projection and stuttering condition are therefore substantive hypotheses, not conventions.

9.2 Passive extensions and conservativity

The passive board extension from the GBG+ reference supplies a section E and a projection U with $U \circ E = \mathrm{id}$ on the underlying GBG data. In that precise passive setting, ordinary game histories are retained and the addition of a distinguished but non-sovereign board-player does not create new ordinary theoremhood. This is a *structural conservativity* result. An active selection or scheduling package may reorganize discovery work, but no runtime improvement follows without a separately analyzed policy and a cost comparison.

9.3 Four logically different settlement predicates

For a target set W in a transition system and an initial state s_0, consider:

Existential reachability:
some legal finite path from s_0 enters W.

Deterministic witness:
a deterministic policy, possibly built from a successful path, reaches W from s_0.

Adversarial forcing:
there exists a strategy for a designated player such that *all* maximal plays consistent with it and all legal opposing choices reach W in finite time.

Certified settlement:
a specified verifier accepts the required proof certificate for the declared target and frozen

hypotheses, with every cost charged according to the protocol.

The first two are equivalent as existence statements for a one-player finite path problem with available legal choices; that equivalence is generally nonuniform, and it need not survive changing the control structure or adding an adversary. Neither implies adversarial forcing. An adversary controlling the choice at a in Figure 1.1 can choose $a \to a$ indefinitely even though b is reachable. Existence of any one of the first three properties does not alone supply the certificate obligations of the fourth.

9.4 A finite forced-win recursion

For a finite turn-based reachability game, let $S_\exists$ be states where the designated controller chooses, and $S_\forall$ states where an opponent chooses. Define $W_0 = T$ and, for each integer $n \geq 0$,

$$\begin{aligned} W_{n+1} = W_n \ &\cup \{s \in S_\exists : \mathrm{Succ}(s) \cap W_n \neq \varnothing\} \\ &\cup \{s \in S_\forall : \mathrm{Succ}(s) \neq \varnothing \text{ and } \mathrm{Succ}(s) \subseteq W_n\}. \end{aligned} \tag{9.1}$$

If all relevant states are finite, this sequence stabilizes, and membership in its limit is the usual reachability-attractor criterion for a forced eventual hit under the stated turn structure and dead-end convention. The proof is an induction on the bounded number of moves needed to force arrival at T. This alternation of existential and universal choices is fundamentally different from a single successful nondeterministic branch, and also different from exact all-history simulation.

9.5 Proof-carrying transport is a separate interface

Let $q : S^+ \to S$ project an extended game to an ordinary proof search. The projected endpoint may coincide with the desired conclusion while the extended certificate relies on rules absent from the ordinary system. A sufficient condition for certificate soundness across the projection is a computable transformer τ such that, for every accepted enlarged proof c^+ of T, the frozen ordinary verifier accepts $\tau(c^+)$ as a proof of T under the unchanged hypotheses. This is a *transport* obligation, not a consequence of state-level projection alone.

9.6 Charged costs and comparison protocols

Any proposed efficiency claim must identify a resource vector, for example

$$C(\pi) = (\#\text{rule attempts}, \#\text{verifier calls}, \text{proof bytes}, \text{peak memory}, \text{elapsed steps}). \tag{9.2}$$

One must fix how the vector is aggregated, whether precomputation and failed searches count, whether input encodings change, and which comparison policy receives the same information. A successful trace preloaded into the selector is advice and must not be credited as free discovery. These are essential logical hygiene conditions whenever search is brought into contact with the realization results.

Chapter 10

From model interfaces to interfaces between theories

10.1 Why simply pooling mathematics is not enough

Two formal theories may speak about different objects even when they use an identical-looking symbol. Before comparing consequences, distinguish their signatures, hypotheses, proof rules, verifiers, and provenance. This issue is unavoidable when abstract probability, proof search, and board-game dynamics are discussed in the same monograph. They are not automatically subtheories of one shared calculus.

Let A_i be the record set for mathematical source i, containing its formulas, assumptions, proofs, verification policy and provenance. The *tagged assembly*

$$Z = \bigsqcup_{i \in I} A_i = \{(i, a) : i \in I,\ a \in A_i\} \tag{10.1}$$

retains the identity of every original record. Each canonical inclusion $\iota_i : A_i \to Z$ has a partial recovery projection $\pi_i : \iota_i(A_i) \to A_i$ satisfying $\pi_i \iota_i = \mathrm{id}_{A_i}$. This elementary assertion is a record-preservation theorem. It does not equate source languages or transfer source theorems to another verifier.

Lemma 10.1 (Lossless recursive assembly)**.** *A finite tree of tagged record unions that retains all original source tags admits, for every leaf i, an inclusion into the root and a partial recovery map on its image whose composite is the identity. Two such trees built from the same tagged leaf records are canonically bijective at the record level.*

Proof. At one node, the inclusion $a \mapsto (i, a)$ is injective and has inverse on its own image. Compose these inclusions and corresponding restricted inverses along the unique path from each leaf to the root. On original leaf tags both assembly trees describe the same disjoint union; the identity on tags and records is the desired bijection. □

10.2 Proof-preserving translations

Write $\Gamma \vdash_i \varphi$ for a derivability judgment in theory T_i and V_i for its fixed proof checker, understood to be sound with respect to that derivability relation. A typed formula translation τ_{ij} is not enough

to preserve proofs. A *proof-preserving interface* additionally requires a computable certificate map κ_{ij} satisfying

$$V_i(c, \Gamma, \varphi) = \text{accept} \quad \Longrightarrow \quad V_j\big(\kappa_{ij}(c), \tau_{ij}(\Gamma), \tau_{ij}(\varphi)\big) = \text{accept}. \tag{10.2}$$

The validity of this implication is a proof obligation about the interface. Without it, one has only a candidate translation or a formal similarity between languages.

Theorem 10.2 (Compositional proof transport)**.** *Suppose (τ_{12}, κ_{12}) and (τ_{23}, κ_{23}) are proof-preserving interfaces, with output signatures, assumptions and certificate formats of the first meeting the inputs of the second. Then the composite $(\tau_{23}\tau_{12}, \kappa_{23}\kappa_{12})$ preserves acceptance of translated derivations.*

Proof. Apply (10.2) to an accepted T_1 certificate to obtain an accepted T_2 certificate. The composability hypothesis allows applying the second transformer to that accepted certificate, yielding an accepted T_3 certificate for the twice translated judgment. The composite transformer is computable because its components are. □

Corollary 10.3 (Certified paths are directed and conditional)**.** *Any finite* directed *chain of composable verified interfaces transports accepted proof judgments along the path. An undirected network of related articles does not, by connectivity alone, supply such transport in either direction.*

10.3 The semantic level: a common reduct

Proof transport is syntactic and checkable. Semantic compatibility is a distinct obligation. Let Σ_1 and Σ_2 be many-sorted first-order signatures meeting precisely in a common signature Σ_0; assume the nonshared symbols are disjoint and the union of theories contains no cross-signature axioms beyond their respective theories. Let $\mathrm{Red}_0(M)$ denote the Σ_0-reduct of a model M.

Proposition 10.4 (Joint-model criterion for disjoint extensions)**.** *Under the preceding assumptions, the union theory has a model exactly when there are models M_1 and M_2 of the component theories whose Σ_0-reducts are isomorphic.*

Proof. Any model of the union restricts to component models with identical, hence isomorphic, common reducts. Conversely transport one component model along an isomorphism of its common reduct so both structures agree on all shared sorts and symbols. Since the other symbols are disjoint, assign their interpretations from their respective component models. Each component's axioms involve only its own signature; consequently the amalgamated structure satisfies both theories. □

Remark 10.5 (Consistency is weaker than an intertwining). The common-reduct condition need not produce an f, R or E satisfying (2.2)–(2.3). An exact dynamical bridge imposes *extra mixed constraints* linking the components. A joint first-order model does not, by itself, imply that the two theories make identical predictions for a measurement or describe the same physics.

10.4 Three interface tests

It is useful to demand that a claimed theory connection pass three separate tests, rather than one vague test:

T1. *Typing and provenance*: every formula, state, assumption and cost belongs unambiguously to a specified source and signature.

T2. *Semantic preservation*: the proposed readout or interpretation preserves the declared observables, laws or admissible histories. A preservation equation is supplied, not merely suggested.

T3. *Certification*: when mathematical claims are transported, a computable certificate transform and the destination verifier's acceptance condition are demonstrated.

For physical comparison a fourth test is usually necessary: the interface must be operationally meaningful under explicitly fixed preparation and measurement permissions. None of these tests follows automatically from a provenance-preserving union of source records.

Chapter 11

A unified hierarchy of histories, choices, and certificates

11.1 The hierarchy

We can now view the results as a hierarchy of *obligations* rather than a hierarchy of philosophical positions. A finite transition graph defines legal state histories. A Markov kernel adds weights to those histories. A hidden deterministic map plus preparation and readout may or may not reproduce the entire resulting joint law. A selection policy traverses just one legal branch. A board-game encoding records legal choices as explicit rules and optional proof witnesses. A board-player may coordinate those rules when authorized and when administrative actions preserve the complete stage-indexed configuration between committed steps. Finally, a proof-translation interface certifies that a claim made in one system survives passage to another.

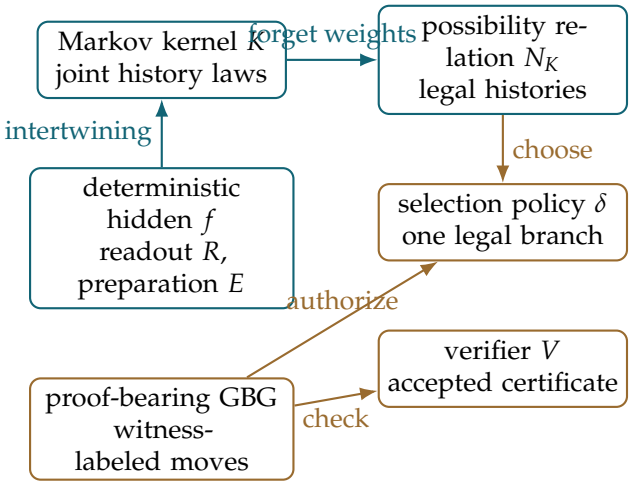


Figure 11.1: Distinct maps do distinct work. The arrows are labeled with the property requiring a definition or proof; they are not assertions that every such arrow exists for arbitrary source systems.

11.2 A precise separation theorem

Theorem 11.1 (One-target deterministic success does not require finite exact realization)**.** *There exists a finite total successor system with a target reachable by a one-step deterministic policy, while no finite hidden deterministic model exactly realizes all its infinite admissible state histories under* (3.1)*. The same support graph admits a Markov weighting for which no finite hidden stochastic interface satisfies* (2.2)–(2.3)*.*

Proof. Take $Y = \{a, b\}$, $N(a) = \{a, b\}$, $N(b) = \{b\}$ and target b. The deterministic selector $a \mapsto b$, $b \mapsto b$ reaches the target in one step. Since a lies on a directed self-loop and has another successor, Theorem 4.3 rules out a finite exact possibilistic realization. Assign probabilities $K(a, a) = K(a, b) = 1/2$, $K(b, b) = 1$ to obtain a stochastic kernel with the same obstructing support graph; the stochastic conclusion follows from the same theorem. □

Proposition 11.2 (Finite-horizon and whole-history requirements diverge)**.** *For any finite-state Markov kernel K with recurrent branching, every $m_h(K)$ is finite and exactly computable by* (5.2)*, but* $\sup_h m_h(K) = \infty$ *and no finite all-time exact realization exists.*

Proof. Each B_{h+1} is finite. Lemma 4.2 says the B_n are unbounded under recurrent branching, and Theorem 5.2 identifies them with m_h. The all-time obstruction is Theorem 4.3. □

11.3 Connections that can be transferred, and those that cannot

The graph-theoretic finite-state theorem applies to a *finite time-homogeneous successor model*. It can be applied to a finite state abstraction of a GBG only after establishing that the abstraction retains the requisite state histories and total successor structure. It is not automatically a theorem about unbounded proof searches with stage-dependent legality. Conversely, the proof-search branch-selection theorem is about *existence of a selected path*, not about exact law reproduction. The two statements can be used side by side precisely because they ask different questions.

Theorem 10.2 also does not transfer the dynamical obstruction into a logical calculus by fiat. To transport it as a theorem, one must specify the source and target formula languages, hypotheses, certificate maps and sound verifiers; the existence of a paper about DNI and another about GBGs is not sufficient. This demand for an explicit proof bridge is the mathematical content of respecting provenance.

11.4 What a physicist may legitimately conclude

The deductions license conditional judgments about finite hidden descriptions, possible trajectory sets, observable path statistics, and costs measured by number of states. They also clarify which kind of search-control guarantee would be needed for an algorithmic application. They do *not* decide whether real-world probabilities are fundamental, whether a continuous hidden-variable theory is viable, or whether deterministic microscale laws make practical prediction possible. Formal determinism, observable stochasticity, probabilistic preparation and computational effectiveness are independent dimensions unless a theorem relates them under declared assumptions.

Chapter 12

Open frontiers, discriminating tests, and nonclaims

12.1 Fixed-source compatibility beyond finite graphs

The finite cardinality classification solves a clean *free realization* problem: choose a hidden model satisfying exact equations, then minimize its number of states. In physics the proposed hidden dynamics, preparation apparatus and readout map may be constrained independently. An appropriate next problem is to fix a class $\mathcal{A}$ of admissible sources (X, f), readouts, preparations, and measurement procedures, and ask whether any member realizes an independently specified target law. The fixed finite model is handled by Proposition 6.1; more general classes need separate mathematics.

Open problem 12.1 (Constrained physical-model compatibility)**.** For a prescribed observable law K, a fixed admissible category $\mathcal{A}$ of source models, and declared operational constraints, decide whether an exact interface exists; when it does, minimize an explicit resource functional $C(X, f, R, E)$. Establish which properties of $\mathcal{A}$ make the decision decidable or yield rigorous obstructions.

The problem is not solved by taking path space without verifying that the path shift belongs to $\mathcal{A}$. Nonclassical and contextual operational theories may require instruments, effects, and composition laws beyond a single finite Markov kernel [10]; invoking them does not invalidate the finite classical theorem but changes its domain.

12.2 Approximate full-history realization

A practical theory should compare entire observed joint distributions on a finite or growing window, not merely the time-n marginal. Let $\mathbb{P}^{(h)}_{K,y}$ denote the Markov law of $(Y_0, \dots, Y_h)$ and $\mathbb{P}^{(h)}_{f,R,E(y)}$ the hidden readout law. One possible resource-sensitive discrepancy is

$$d_h = \sup_{y \in Y} \left\| \mathbb{P}^{(h)}_{K,y} - \mathbb{P}^{(h)}_{f,R,E(y)} \right\|_{\mathrm{TV}}. \tag{12.1}$$

Research could minimize $|X|$ or actual circuit cost subject to $d_h \leq \varepsilon$. This would interpolate between exact horizon realizability and physically relevant approximation. The marginal bound of Theorem 6.2 should not be substituted for an unproved estimate of d_h.

Open problem 12.2 (Approximation–memory tradeoff)**.** For finite K, specified h and ε, determine the minimum finite hidden-state size under $d_h \leq \varepsilon$, and characterize the scaling with h, ε and the support structure. Do not presume that support combinatorics alone determines the answer once zero-probability events may be approximated.

12.3 Effective control rather than supplied winning advice

Theorem 7.2 turns a given successful trace into a deterministic witness policy. A research-grade efficiency claim requires *finding* such a trace algorithmically, with fully charged costs. It is reasonable to investigate fair scheduling, reversible suspension of unproductive approaches, verifier-gated lemmas and controlled memory. None of those mechanisms is established here to provide a general speedup, and their proof obligations cannot be discharged by structural analogies.

Open problem 12.3 (Discoverable certified control)**.** Give structural conditions on a search graph and fixed verifier under which an algorithm produces a successful selector, proves its certificate correctness, and beats a declared comparator under a fully specified end-to-end resource measure. Separate existence of a good selector from a uniform efficient procedure to construct it.

12.4 Interfaces that preserve only goal-relevant behavior

Whole-history realization may be excessive for practical planning. A target predicate $W \subseteq S$ suggests weaker equivalence classes of histories, for example equivalence by *whether* and *when* W is first reached. In some settings a tiny finite controller may preserve this target statistic although the complete history process requires unbounded hidden resources. But preserving reachability or first-hit distributions requires its own equations, and a claim about one target must not be rephrased as equivalence of entire theories.

Open problem 12.4 (Goal-sensitive interface complexity)**.** For a finite Markov or possibilistic process, a target set W, and a declared class of target observations, determine the minimum hidden resources for exact preservation of target behavior. Characterize conditions under which target-sensitive costs are strictly smaller than B_{h+1} without conflating deterministic selection with stochastic first-hit distributions.

12.5 Proof-theoretic and semantic amalgamation

Beyond elementary shared-signature compatibility, a genuine interface between scientific theories requires a map of observables, models, satisfaction and allowed experiments. Institution theory [11] offers a standard framework for logic translations; Markov categories [9] address stochastic compositional structure. An ambitious general theory must state which pieces are preserved and which are not, and supply witnesses. No single categorical diagram eliminates the separate certificate and physical-model constraints.

Open problem 12.5 (Certified common-theory synthesis)**.** Given two independently specified formal/operational theories, a shared observational fragment and prescribed allowable translations, find necessary and sufficient conditions for a common model *with exact mixed interface constraints*, and construct independently acceptable proof certificates for any transported results. A purely tagged union or an isomorphism of common reducts is not a solution.

12.6 A negative-results discipline

A theorem may be strong precisely because its nonclaims are sharp. We record five boundaries that no later speculative discussion should silently remove:

1. No finite exact representation does *not* imply irreducibly stochastic physics.
2. A single successful deterministic search trace does *not* imply efficient uniform discovery.
3. Equal successor graphs do *not* imply equal probabilistic behavior.
4. Structural inclusion of papers or theories does *not* imply certified theorem transport.
5. A verifier-controlled target reachability theorem does *not* imply adversarial forced victory or a measured computational improvement.

These boundaries are part of the mathematics, not caveats appended after the results. They delimit what future experiments, applications and philosophical interpretations would have to establish.

12.7 Conclusion

The constructive message is precise. Deterministic hidden dynamics can reproduce stochastic observations when preparation and readout satisfy an exact intertwining; for finite classical systems, recurrent branching is the sharp obstruction to doing so with finitely many hidden states for all time. For every fixed observational horizon the obstruction disappears, and the exact minimum finite memory is the number of legal observable words. Nondeterministic proof search obeys another logic: legal possibilities can be encoded as explicit rules, while reaching one target requires selecting only one successful branch, not reconstructing the full possibility space. Certified control and translation bring their own witness, authorization and verification requirements. These distinctions produce useful theorems without forcing any position on the ontology of physical chance.

Appendix A

Notation, assumptions, and compact proof checklist

A.1 Glossary of primary objects

Symbol or term	Precise meaning
Y	Finite nonempty observable state set (Chapters 2–6), or a formula domain when explicitly redefined (Chapters 7–9).
X	Hidden-state set, required finite in finite-resource results.
$K(y, z)$	Row-stochastic Markov transition probability.
$N_K(y)$	Set of successors with strictly positive $K(y, z)$.
$f : X \to X$	Deterministic hidden-time evolution, fixed after preparation.
$R : X \to Y$	Observable readout map.
$E(y)$	Preparation law for hidden state conditional on observable y.
$L(y)$	Set of hidden states admissible for initial observable y.
$\mathrm{Word}_n(N)$	Legal *state* words of length n, counting all initial states.
$B_n(N)$	Cardinality of $\mathrm{Word}_n(N)$.
$\mathrm{Path}_\infty(N)$	Set of legal infinite state sequences.
$m_h(K)$	Least hidden cardinality for exact joint laws of the first $h + 1$ observations; no all-time intertwining imposed.
$m_\infty(K)$	Least *finite* hidden cardinality for an all-time interface; undefined as a finite minimum in obstructed cases.
$\Sigma(p, m)$	Nonempty set of admissible finite proof-state extensions at stage m.
$\mathcal{W}(p, m, \Delta)$	Set of accepted finite witness encodings for an extension.
$P_0 = B^\star$	GBG+ distinguished board-player with only declared powers.
τ, κ, V	Typed formula translation, certificate transformer, and declared verifier.

A.2 Which result needs which hypothesis?

Result	Irreducible premises or restrictions
All-history Markov law	Measurable f, R, E, finite Y, exact (2.2)–(2.3).

Result	Irreducible premises or restrictions
Support bridge	Finite X, Y, nonnegative stochastic weights, exact intertwining.
Infinite finite-state dichotomy	Finite *total time-homogeneous* positive-transition graph; freely chosen source f, R, E; full history preservation.
Sharp horizon minimum	Finite Y, row-stochastic K, exact joint-history matching through *fixed* horizon h; freely chosen finite X.
Fixed-model LP	Finite, already prescribed f, R, K; only preparation probabilities variable.
Marginal TV error	Exact initial readout; uniform one-step TV interface error; single-time output comparison.
Deterministic trace selector	Finite hypothesis set and *given finite successful branch*, stage-complete states, accepted witnesses on its transitions; effective off-trace choice needed for executable selection. Numerical stage bounds require explicit indexing conventions.
GBG compilation	Finite encodings, witness coverage for every admitted extension, genuine consequence witnesses; executable compilation additionally needs decidable legality and a total sound verifier.
GBG+ controller simulation	Explicit selection/commit authority, stage-complete projection $q(s) = (p, m)$, administrative q-stuttering, one increment per commitment, finite blocks, and frozen-verifier soundness for trusted deposits.
Proof transport	Well-typed formula and certificate translations; destination-verifier acceptance; composable hypotheses.
Joint-model compatibility	Disjoint additional signatures, common reducts isomorphic, no cross-axioms.

A.3 A simple hand computation of B_n

For the absorbing graph with order (a, b),

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad A^h = \begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{1}^\mathsf{T} A^h \mathbf{1} = h + 2.$$

For the complete binary graph,

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad B_{h+1} = 2^{h+1}.$$

These are direct tests of the exact finite-horizon formula. In either graph the infinite-history criterion fails because a cycle vertex branches. The contrast is a transparent counterexample to an unwarranted assertion that recurrent branching always forces exponential state growth.

A.4 A note about horizons and terminal states

A terminal state has no outgoing edge. The all-time definitions in this monograph require a total successor relation, so a terminal state may be equipped with an explicit absorbing self-loop to represent continued observation after termination. This is a *modeling choice* which changes the set of infinite histories while retaining the intended finite reachability of the terminal target. When comparing move-labeled games, adding an absorbing transition also changes move histories unless the added terminal stutter is treated as a distinguished administrative convention. Such modifications must be declared.

Appendix B

Source integration and evidentiary ledger

B.1 The four DNI-related documents and their mathematical roles

This monograph integrates, rather than juxtaposes, four previously prepared DNI-related documents:

1. *Deterministic–Nondeterministic Interfaces*, ATLAS-repaired working note v0.2 [1]: source for strict stochastic and possibilistic intertwining, the positive-support bridge, the finite all-time obstruction, fixed-implementation linear feasibility, and the marginal error estimate.
2. *Zip–Inverse-Zip Interface Theory* [2]: source for tagged record preservation, partial inverse projections, typed certificate transformations and their composability. These are moved here to the broader chapter on logic and semantic compatibility, not inserted as a detached article.
3. *Deterministic–Nondeterministic Interfaces: A Textbook-Ordered Reader* [3]: an editorial collation of the previous two mathematical pieces and an internal audit. It is one of the four requested documents but contributes no additional independent theorem source or separate numerical credit.
4. *Search, Control, and Realization*, repaired v0.2 [4]: source for the rigorous separation of selected target trajectories from full-history realization, the sharp $m_h = B_{h+1}$ finite-horizon theorem, and witness-complete encoding of nondeterministic proof search into board games. Its arguments are recast in Chapters 5 and 7–9.

These pieces connect to, but do not contain in their entirety, two further source theories: *Depths of Induction* v49 [5] supplies the nondeterministic search-policy definitions and proof-covering perspective; the GBG/GBG+ foundation [6] supplies the formal move-rule dictionary and the authorized board-player. Their relevant assumptions have been restated where used. This manuscript does not claim to reproduce the entirety of either work.

B.2 What is genuinely integrated here?

The monograph introduces a single terminology and mathematical dependency order: (i) specify observable histories and preservation equations, (ii) prove exact realizability and resource obstructions, (iii) explain weaker finite-horizon preservation, (iv) separate deterministic choice from all-history realization, (v) model witness-labeled choice by legal formal moves, and (vi) require

certified transport between independent theories. This is a new presentation of the material with cross-references and explicit boundaries; the earlier papers' established results are attributed, not relabeled as fresh discoveries.

B.3 Evidence categories

Classical antecedents.
Markov intertwinings, path shifts, basic directed-graph arguments, and standard logical-model amalgamation are established mathematical constructs. Claims of historical priority are not made without an independent literature investigation.

Self-contained derivations.
The finite all-history criterion, exact horizon minimum, recurrence and LP feasibility, branch-selection and witness-compilation claims are supported in this monograph by explicit mathematical proofs under displayed hypotheses.

Conditional architecture.
The active GBG+ controller result holds under declared legal moves, full stage-preserving projection and administrative stuttering, finite commitment blocks, and verifier rules. Existence of those packages in a particular implementation is not presumed.

Unsolved frontier.
Uniform efficient discovery, fixed physical source compatibility, cost-optimal certified settlement, a universal interface of physical theories, and operationally justified quantum or other nonclassical extensions are not established by the present results.

B.4 Repair provenance for the current edition

Version 0.2 retains the finite-state and finite-horizon mathematical core of Version 0.1. In response to the ATLAS monograph audit, the controller simulation now requires preservation of the complete stage-indexed state during administrative actions and an exact increment on committed actions. A new effective-compilation corollary separates finite encodings and abstract witness existence from decidable legality and executable verification. The all-history discussion now explicitly distinguishes permitted infinite paths from positive-probability singleton paths. The updated assumptions are mirrored in the checklist in Appendix A. No claim of a general search speedup, physical hidden-variable ontology, machine-checked proof, or external refereeing is added.

B.5 Why this is a research monograph rather than a collected reader

The mathematical story here is not organized by which earlier document happened to introduce a formula. It is organized by one research problem: *which properties of dynamical and inferential systems survive a specified interface, and what resources are necessary?* Definitions and proofs are positioned at the point at which the argument requires them. Examples serve as discriminating countermodels, not anecdotes. Original source papers and the editorial reader are identified as prior materials; they are neither merely merged into one file nor promoted by association to a

universal theory. This authorial separation also respects the roles of the curator and interlocutor, who is not the author of this monograph.

Bibliography and source provenance

[1] B. Tenneson, *Deterministic–Nondeterministic Interfaces: Exact Realization, Finite-Resource Obstructions, and Common-Theory Compatibility*, ATLAS-repaired mathematical working note, v0.2, October 2026. Primary finite classical source; proofs and scope statements treated as working research.

[2] *Zip–Inverse-Zip Interface Theory: Structural Recovery, Certified Transport, and Compositional Interfaces*, research exposition, October 2026. Structural provenance and conditional proof transport.

[3] *Deterministic–Nondeterministic Interfaces: A Textbook-Ordered Reader of Core Results, Zip–Inverse-Zip Structures, and Mathematical Audit*, collected research materials, October 2026. An editorial container of [1, 2] and an internal audit, not a distinct independent theorem source. https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/deterministic_nondeterministic_interfaces_textbook_ordered_reader_20261020/.

[4] DATA MIND, *Search, Control, and Realization: A Three-Article Multiarticle Theory of Deterministic and Nondeterministic Proof Search on Generalized Board Games*, internally ATLAS-repaired synthesis, v0.2, October 2026. The exact finite-horizon bound and the witness-complete search compilation are integrated and re-proved in this monograph.

[5] B. Tenneson, *Depths of Induction and Formalized Self-Awareness*, v49, 2026, especially Section 5 on nondeterministic search policies. https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/depths_of_induction_v49/.

[6] B. Tenneson, *Generalized Board Games and GBG+: A Self-Contained Formal Foundation with the Board as a Player*, September 2026, especially Sections 4–11 and 16. https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/gbg_and_gbg_plus_formal_foundations_004/.

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[11] J. A. Goguen and R. M. Burstall, "Institutions: Abstract model theory for specification and programming," *Journal of the ACM* **39** (1992), 95–146.

Method and status

This document was newly organized and written by DATA MIND as an AI-authored mathematical research monograph. Brian Tenneson served as curator and interlocutor and is not an author of this document; the underlying cited papers retain their original authorship. The arguments in this manuscript are mathematical derivations and constructions presented for critical audit; no external referee report or machine-checked proof certificate is claimed. The ATLAS review of monograph v0.1 identified a stage-projection hypothesis requiring repair and recommended a sharper separation of mathematical witnesses from executable verification. Both changes are incorporated in this edition. Any later internal audit of this version is separate from independent peer review or proof-assistant verification. No prover, physical simulation, settlement experiment, or new data collection was conducted in preparing it.